

$$f(x) = \sqrt[4]{x^2 - 7x + 10}$$

$$f(x) = \sqrt[4]{x^2 - 7x + 10}$$

□ D(f): $x^2 - 7x + 10 \geq 0$

$$(x-2)(x-5) \geq 0$$

$$\overline{+} - \overline{+} \Rightarrow$$

$$\Rightarrow D(f) = (-\infty; 2] \cup [5; \infty)$$

$$f(x) = x^2 + \tan(x)$$

$$f(x) = x^2 + \tan(x)$$

$$D_1(x^3) = (-v, v)$$

$$D_1(t, x) \in R \setminus \left\{ \frac{\pi}{2} + k\pi \right\}, \quad k \in \mathbb{Z}$$

$$\mathcal{D}(A) = P_1 \cap P_2 = \mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi; k \in \mathbb{Z} \right\}$$

$$f(7) = \sqrt{7-7} + \sqrt{10-7} = 0 + \sqrt{3} = \sqrt{3}$$

$$f(7) = \sqrt{7-7} + \sqrt{10-7} = 0 + \sqrt{3} = \sqrt{3}$$

$$D_1(\sqrt{71}-7) \Rightarrow 71-7 \neq 0 \text{ WJA}$$

$$7 \rightarrow 7 \Rightarrow D_1 = (\sqrt{7-7}) = [7, 0)$$

Q/3 6.1.3-6.1.22

6.1.3
 $f(x) = \frac{x^2+4}{x^2+1}$

□

$$D(f): x^2+1 \neq 0$$

$$x^2 \neq -1$$

$$x \neq \pm i$$

$$\Rightarrow D(f) = (-\infty; -i) \cup (-i; i) \cup (i; \infty)$$

6.1.4

$$f(x) = \sin x \cdot \frac{1}{|x|-2}$$

□

$$D_1(\sin x) = (-\infty; \infty), D_2\left(\frac{1}{|x|-2}\right) \Rightarrow |x|-2 \neq 0$$

$$x \neq \pm 2$$

$$D_1(x) = (-\infty; \infty)$$

$$D(f) = (-\infty; -2) \cup (-2; 2) \cup (2; \infty)$$

6.1.5

~~11/1~~ $f(x) = \log_3(-x)$

□

$$D(f): -x > 0$$

$$x < 0 \Rightarrow D(f) = (-\infty; 0)$$

$$D_2(\sqrt{10-x}): 10-x \geq 0 \\ x \leq 10 \Rightarrow D_2 = (-\infty; 10]$$

$$D(f) = D_1 \cap D_2 = [7; 10]$$

b.1.2

$$f(x) = \frac{\ln(x)}{\sqrt{|x^2-2|}}$$

□

$$D_1(\ln(x)) = x > 0 \Rightarrow D_1 = (0; \infty)$$

$$D_2(\sqrt{|x^2-2|}): |x^2-2| \geq 0 \\ x \in \mathbb{R} \Rightarrow D_2 = (-\infty; \infty)$$

$$D_3 = \left(\frac{1}{\sqrt{|x^2-2|}} \right) \neq 0$$

$$|x^2-2| \neq 0$$

$$x^2-2 \neq 0$$

$$x^2 \neq 2$$

$$x \neq \pm\sqrt{2} \Rightarrow D_3 = (-\infty; -\sqrt{2}) \cup$$

$$\cup (-\sqrt{2}; \sqrt{2}) \cup (\sqrt{2}; \infty)$$

$$D(f) = (0; \sqrt{2}) \cup (\sqrt{2}; \infty)$$

6.1.15

$$f(x) = 3^{-x^2}$$

□

~~lim~~

$$3^0 = \frac{1}{3^0} = E(3^{x^2}) = [1; 1]$$

$$E\left(\frac{1}{3^x}\right) = \left[\frac{1}{1}, \frac{1}{\infty}\right) = [1; 0)$$

$$\Rightarrow E(3^{-x^2}) = (0, 1]$$

6.1.16

$$f(x) = 2 \cdot \sin x - 7$$

□

$$E(\sin x) = [-1; 1]$$

$$E(2 \sin x) = [-2; 2]$$

$$E(2 \sin x - 7) = [-2-7; 2-7] =$$

$$= [-9; -5]$$

6.1.20

$$f(x) = \frac{x+3}{x^2-1}$$

□

$$1) f(0) = \frac{0+3}{0^2-1} = \frac{3}{-1} = -3$$

6.1.10

$$f(x) = \sqrt[4]{x+2} + \frac{1}{\sqrt{1-x}}$$

□

$$D_1 = (\sqrt[4]{x+2}) \Rightarrow x+2 \geq 0$$

$$x \geq -2 \Rightarrow D_1 = [-2; \infty)$$

$$D_2 = \left(\frac{1}{\sqrt{1-x}} \right) \Rightarrow \frac{1}{\sqrt{1-x}} \neq 0$$

$$1-x \neq 0$$

$$x \neq 1 \Rightarrow D_2 = (-\infty; 1)$$

$$D(f) = [-2; 1)$$

6.1.14

$$f(x) = x^2 - 8x + 20$$

□

$$\begin{aligned} f(x) = x^2 - 8x + 20 &= (x^2 - 8x + 16) - 16 + 20 = \\ &= (x-4)^2 + 4 \end{aligned}$$

$$(x-4)^2 \geq -4 \Rightarrow [-4; 8)$$

$$E(f) = [-4; \infty)$$

$$2) f(-2) = \frac{-2+3}{(-2)^2-1} = \frac{1}{4-1} = \frac{1}{3}$$

$$3) f(\sqrt{2}) = \frac{\sqrt{2}+3}{(\sqrt{2})^2-1} = \frac{\sqrt{2}+3}{2-1} = \sqrt{2}+3$$

$$4) f(-x) = \frac{-x+3}{(-x)^2-1} = \frac{-x+3}{x^2-1}$$

$$5) f\left(\frac{1}{x}\right) = \frac{\frac{1}{x}+3}{\left(\frac{1}{x}\right)^2-1} = \frac{3+\frac{1}{x}}{\frac{1}{x^2}-1}$$

$$6) f(u+1) = \frac{u+1+3}{(u+1)^2-1} = \frac{u+4}{u^2+2u+1-1} = \frac{u+4}{u^2+2u}$$

$$7) f(u) = \frac{u+3}{u^2-1} + 1 = \frac{u+3+u^2-1}{u^2-1} = \frac{u^2+u+2}{u^2-1}$$

$$8) f(x^2) = \frac{x^2+3}{(x^2)^2-1} = \frac{x^2+3}{x^4-1}$$

6.1.21

$$f(x) = 7(3 \cdot 2^x)$$

□

$$1) f(1) = 13 \cdot 2^1 = 26$$

$$2) f(-3) = (-3)^3 \cdot 2^{-3} = -\frac{27}{8}$$

$$3) f(-\sqrt[3]{5}) = (-\sqrt[3]{5})^3 \cdot 2^{-\sqrt[3]{5}} = -\frac{5}{2^{\sqrt[3]{5}}}$$

$$4) \varphi(z+3) = \frac{\sqrt{z+3+5}}{(z+3)^2} = \frac{\sqrt{z+8}}{z^2+6z+9}$$

$$5) \varphi(z+1) = \frac{\sqrt{z+1+5}}{(z+1)^2} = \frac{\sqrt{z+6}}{z^2+2z+1}$$

$$(f)_{20} = (f)$$

$$f_{20} = f$$

$$\frac{f}{f} = \frac{f_{20}}{f} = 1 = \frac{f}{f} \cdot 20$$

$$(20) = (f) + 20$$

$$\uparrow 0 < x \text{ nur}$$

$$\downarrow 0 > x \text{ nur}$$

$$E(\varphi(z+1)) = 20 \cdot f = 20 \cdot f$$

$$(1 - K f) \cdot f = (K f) \cdot f$$

$$= (1 - K f) \cdot f = (K f) \cdot f$$

$$(1 - \frac{f}{f} + K f) \cdot f =$$

$$\frac{f}{f} = 1 \cdot (1 - K f) \cdot f$$

$$(1 - K f) \cdot f = (\frac{f}{f}) \cdot f = (K f) \cdot f$$

$$K = \frac{f}{f} = 1 \cdot \frac{f}{f} \cdot f$$

$$K = 1 \cdot f = f$$

$$4) f(-x) = (-x)^3 \cdot 2^{-x} = -x^3 \cdot \frac{1}{2^x} = -\frac{x^3}{2^x}$$

$$5) f(3x) = (3x)^3 \cdot 2^{3x} = 27x^3 \cdot 2^{3x} = 27x^3 \cdot 8^x$$

$$6) f\left(\frac{1}{x}\right) = \left(\frac{1}{x}\right)^3 \cdot 2^{\frac{1}{x}} = \frac{2^{\frac{1}{x}}}{x^3}$$

$$7) \frac{1}{f(x)} = \frac{1}{x^3 \cdot 2^x} = (x^3 \cdot 2^x)^{-1}$$

$$8) f(8-2) = (8-2)^3 \cdot 2^{8-2} = (8^3 - 6 \cdot 8^2 + 12 \cdot 8 - 8) \cdot 2^6$$

$$= 2^{10}$$

6.1.22

$$\varphi(t) = \frac{\sqrt{t+5}}{t^2}$$

$$1) \varphi(-1) = \frac{\sqrt{-1+5}}{(-1)^2} = \frac{\sqrt{4}}{1} = 2$$

$$2) \varphi(-9) = \frac{\sqrt{-9+5}}{(-9)^2} = \frac{\sqrt{4}}{81} = 0$$

$$3) \varphi\left(\frac{5}{4}\right) = \frac{\sqrt{\frac{5}{4}+5}}{\left(\frac{5}{4}\right)^2} = \frac{\sqrt{\frac{25}{4}}}{\frac{25}{16}} = \frac{\frac{5}{2}}{\frac{25}{16}} = \frac{5}{5} = 1$$

$$= \frac{8}{5}$$

Р/3 6.1.26-6.1.38

6.1.26

□

1) $f(x) = \cos(\frac{x}{4})$

$\cos x: T = 2\pi$

$\cos \frac{x}{4}: T = \frac{2\pi}{\frac{1}{4}} = \frac{\pi}{2}$

2) $f(x) = |x|$

$x \geq 0 \uparrow$

$x < 0 \downarrow$

$[0; \infty) = \forall x \in (-\infty; \infty) \exists$

3) $f(x) = \tan(2x-1)$

$\tan(2x-1) = \tan(2x+\pi-1) =$

$= \tan(2(x+\frac{\pi}{2})-1)$

$\tan(2x-1): T = \frac{\pi}{2}$

4) $f(x) = \sin(\frac{x}{2}) = \cos(\frac{x}{2} + \frac{\pi}{2})$

$\sin(\frac{x}{2}): T = \frac{2\pi}{\frac{1}{2}} = \pi$

$\cos x: T = \pi$

$$\sin\left(\frac{\pi}{2} + T_1\right) - \cos\left(\frac{\pi}{2} + T_1\right)$$

$$\text{HOK}(T_1; T_2) = \text{HOK}(\pi; \pi) = \pi$$

$$5) f(x) = \sin(3x) \cdot \cos(3x)$$

$$\sin(3x); T = \frac{2\pi}{3}$$

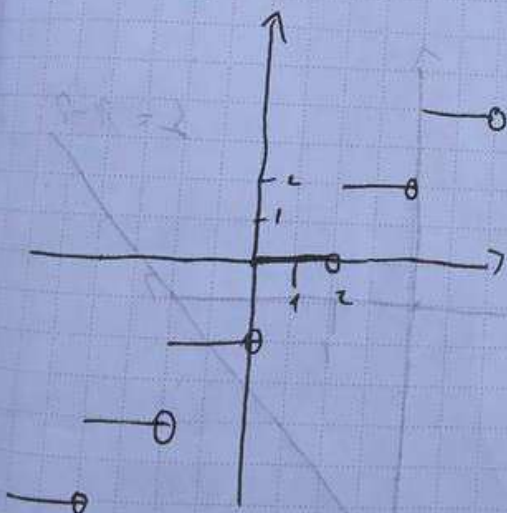
$$\cos(3x); T = \frac{2\pi}{3}$$

$$\text{HOK}(T_1; T_2) = \text{HOK}\left(\frac{2\pi}{3}; \frac{2\pi}{3}\right) = \frac{2\pi}{3}$$

6.1.27

4

$$1) L = [7; 1]$$



$$(x-1)(x-2)$$



6.1.41

$$y = 3x + 5$$

□ $\text{na } (-\infty; \infty) f(x) \uparrow$

$$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2) \Rightarrow$$

$\exists$.

$$x = \frac{y-5}{3} \Rightarrow y^{-1} = \frac{x-5}{3}$$

6.1.42

$$y = x^3 - 2$$

□ $\text{na } (-\infty; \infty) f(x) \uparrow, x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2) \Rightarrow$

$$x \neq \sqrt[3]{y+2} \Rightarrow y^{-1} = \sqrt[3]{x+2}$$

6.1.43

$$y = |x|$$

□

$\text{na } [-a; 0] f(x) \downarrow, (0; a] \uparrow; \forall \varepsilon > 0; \exists \delta > 0; \exists x_1 = -\varepsilon, x_2 = \varepsilon$

$$\nexists y^{-1}$$

6.1.44

$$y = \frac{x-2}{x}$$

□

$D(f); x \in \mathbb{R} \setminus \{0\}$

на $(-\infty, \infty)$ $f(x)$, т.о. $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2) \Rightarrow$

$$x = \frac{y-2}{y}; \quad x \neq \frac{y-2}{y}; \quad y - x \neq 2; \quad y - (1-x) \neq 2 \Rightarrow$$

$$\Rightarrow y^{-1} = \frac{2}{1-x}$$

6.1.45

$$f(x) = \lfloor$$

□ $f(x)$ монотонна $f(x)$ не $\uparrow$ и не $\downarrow$
 $f(x)$ ограничена.

6.1.46

$$f(x) = \sin^2 x$$

□ Ограничена $f(x)$ $[-1, 1]$

6.1.47

$$f(x) = \arctan x$$

□ Строго-монотонная $(f(x), \uparrow)$
 Ограничена $[-\frac{\pi}{2}, \frac{\pi}{2}]$

6.1.48

$$f(x) = -x^2 + 2x$$

□

не убн, не максимизируй, не строго монот, не
ограничен на $(-\infty; 1)$ $f(x) \uparrow$ на $[1; \infty)$

6.1.49

$$f(x) = \frac{x+2}{x+5}$$

□

строгo монотонно сп-я на $(-\infty; \infty)$ $f(x) \uparrow$

6.1.54

□

$$1) \frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$\frac{y^2}{4} = \frac{x^2}{9} - 1$$

$$y^2 = \frac{4x^2}{9} - 4$$

$$y = \pm \sqrt{\frac{4x^2}{9} - 4}$$

$$y = \pm \sqrt{\frac{4x^2 - 36}{9}}$$

$$y = \pm \frac{2}{3} \sqrt{x^2 - 9}$$

$$2) \lambda(1+y) = 1$$

$$|y| = 1 - \lambda$$

$$\lambda = \pm(1-\lambda) \quad \text{т.к. } \lambda \leq 1$$

$$3) e^y - \sin y = x^2$$

6.1.55

$$y + \cos(y) - x = 0$$

□

$$1) A(1; 0)$$

$$0 + \cos(0) - 1 = 0$$

$$0 = 0 \Rightarrow \text{нормально}$$

$$2) B(0; 0)$$

$$0 + \cos(0) - 0 = 1$$

$$1 \neq 0 \Rightarrow \text{нет}$$

$$3) \left(\frac{\pi}{2}; \frac{\pi}{2}\right)$$

$$\frac{\pi}{2} + \cos \frac{\pi}{2} - \frac{\pi}{2} = 0$$

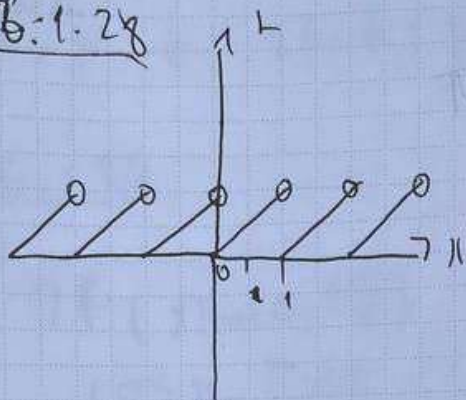
$$0 = 0 \Rightarrow \text{нормально}$$

$$4) D(\pi-1; \pi)$$

$$\pi + \cos(\pi) - \pi + 1 = 0$$

$$0 = 0 \Rightarrow \text{нормально}$$

6.1.28

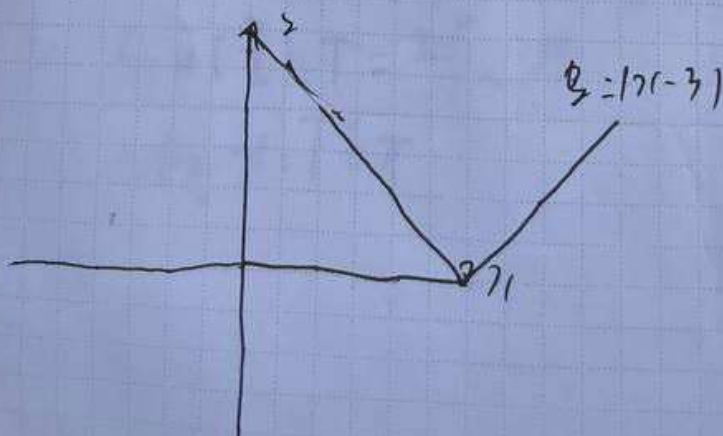
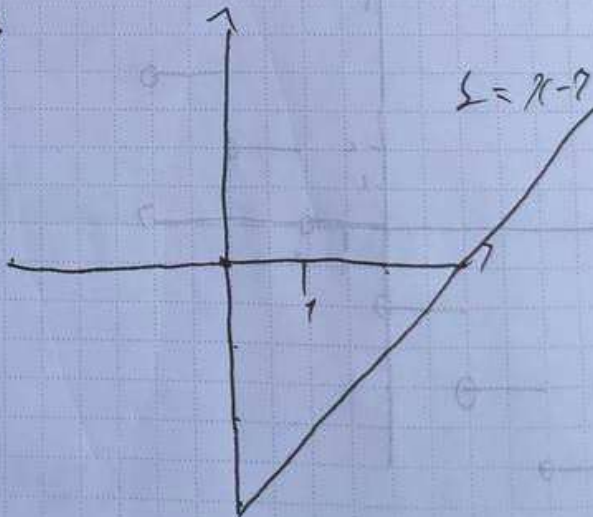
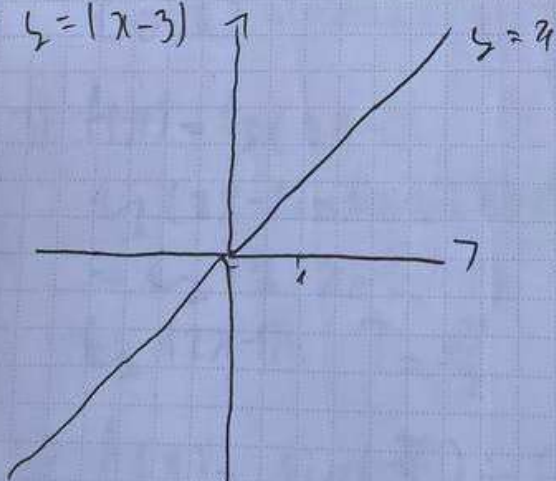


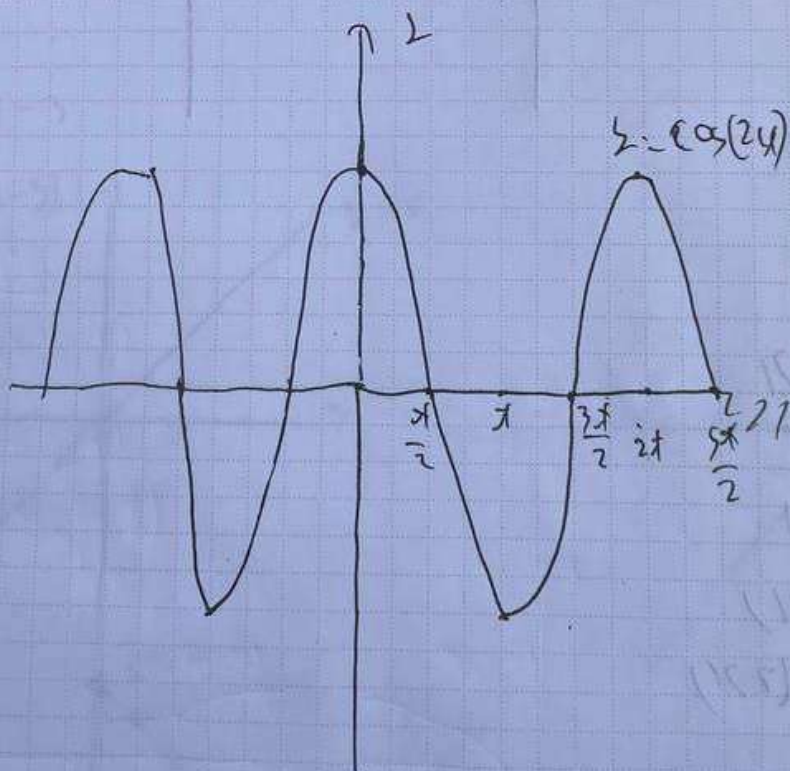
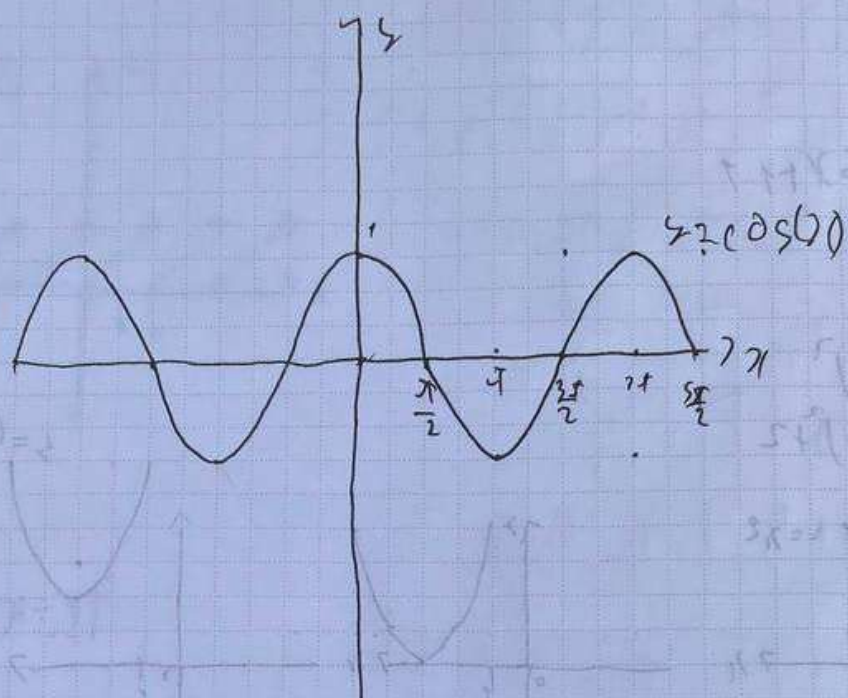
6.1.29

$$y = |x-3|$$

□

- $y = x$
- $y = x-3$
- $y = |x-3|$





6.1.30

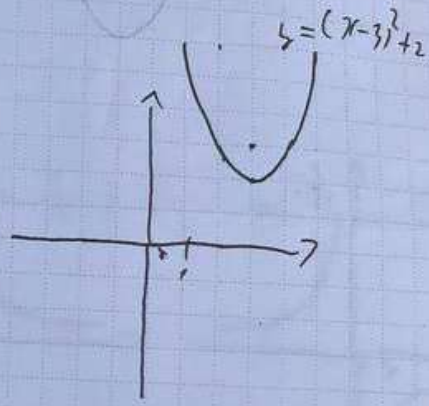
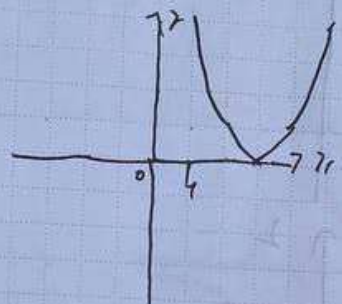
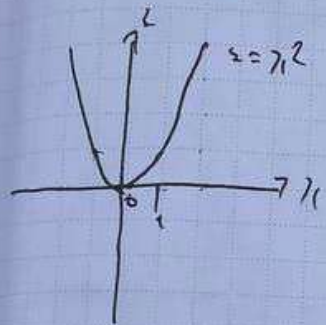
$$y = x^2 - 6x + 11$$

□

$$y = x^2$$

$$y = (x-3)^2$$

$$y = (x-3)^2 + 2$$



6.1.31

$$y = 3 \cos(2x)$$

□

$$y = \cos(x)$$

$$y = \cos(2x)$$

$$y = 3 \cos(2x)$$

6.1.33

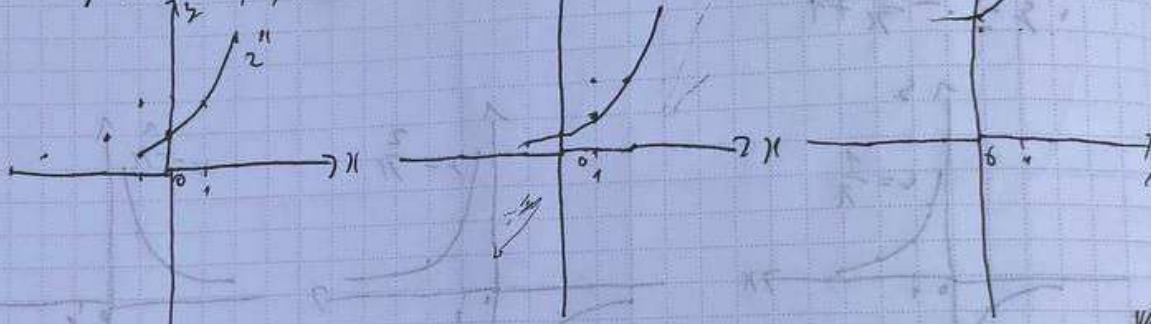
$$y = 2^{x-1} + 3$$

□

$$y = 2^{1(x-1)}$$

$$y = 2^{1(x-1)}$$

$$y = 2^{x-1} + 3$$



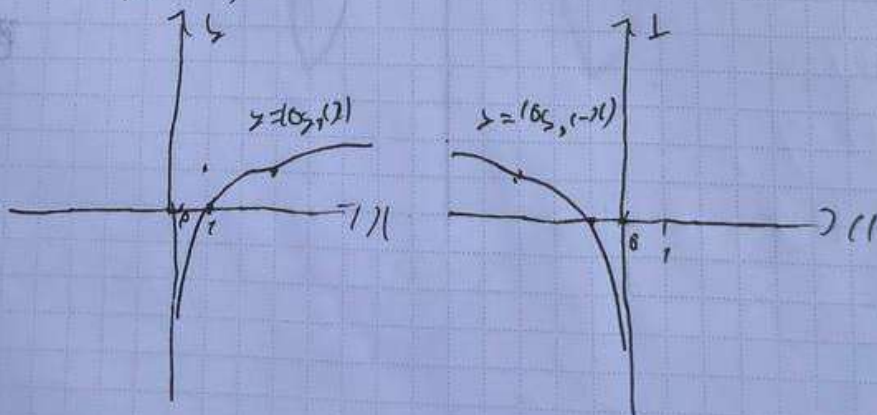
6.1.34

$$y = \log_2(-x)$$

□

$$y = \log_2(x)$$

$$y = \log_2(-x)$$



6. 1.32

$$v = -\frac{2}{\pi} + 1$$

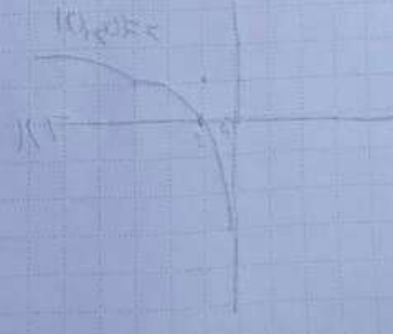
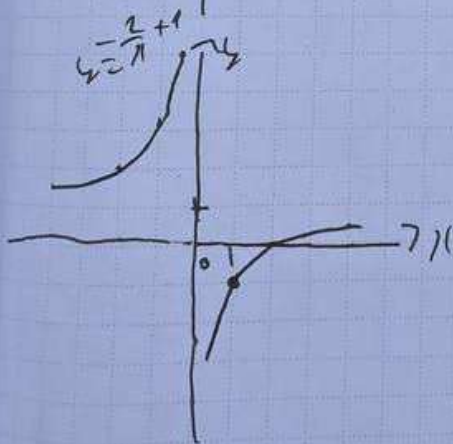
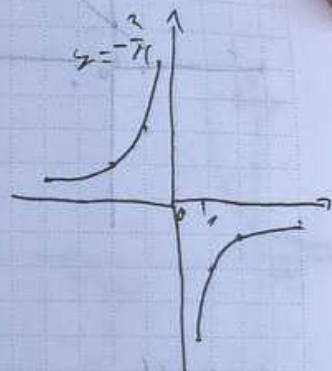
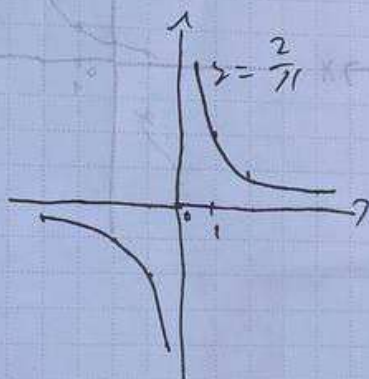
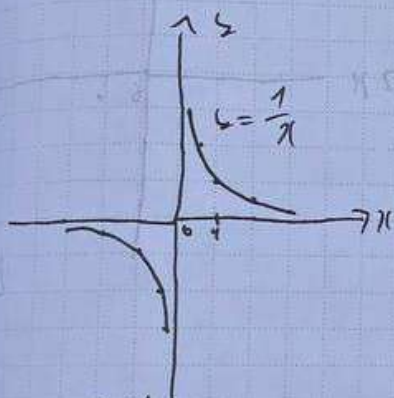
□

$$\cdot \gamma = \frac{1}{\pi}$$

$$\cdot \gamma = \frac{2}{\pi}$$

$$\cdot \gamma = -\frac{2}{\pi}$$

$$\cdot \gamma = -\frac{2}{\pi} + 1$$



6.1. 36

$$y = \frac{x+4}{x+2}$$

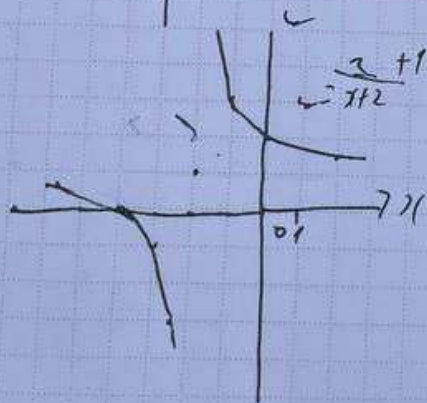
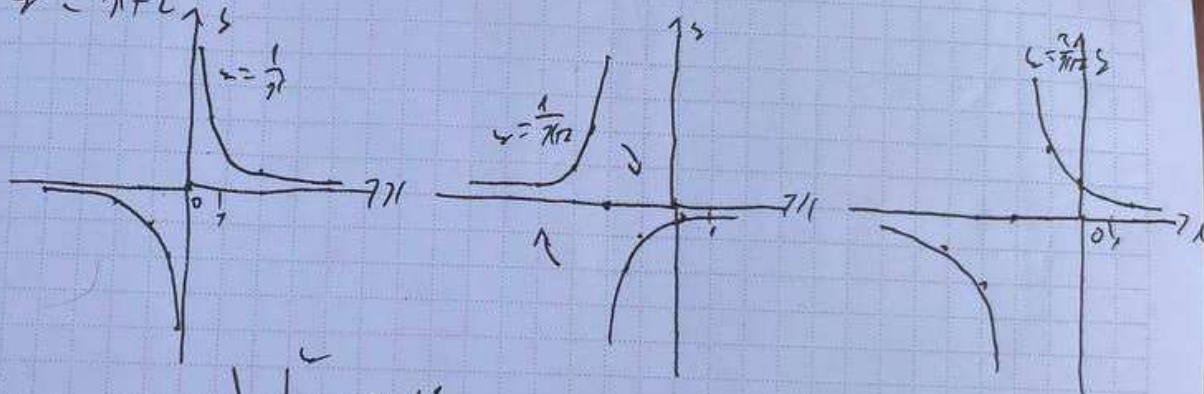
□

$$y = \frac{1}{x}$$

$$y = \frac{1}{x+2}$$

$$y = \frac{2}{x+2}$$

$$y = \frac{2}{x+2} + 1$$



$$\begin{pmatrix} 1 & \frac{1}{3} \\ 2 & \frac{1}{3} \end{pmatrix}$$